Fourier Series and Partial Differential Equations

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from reading the harmonics of a signal and computing Euler–Fourier coefficients, through even/odd symmetry and half-range expansions, into separation of variables and the Laplacian, and up to a complete separation-of-variables solution of the heat equation with every coefficient integral expanded, plus the d'Alembert solution of the wave equation. Solutions follow in Part III.

1. **(Harmonics)** For $f(x) = 4 - 3\cos(2\pi x) + \sin(\pi x)$ with period 2 ($L = 1$), state a_0, the nonzero a_n, and the nonzero b_n.
2. **(Euler–Fourier setup)** Write the Euler–Fourier formulas for a_n and b_n of a $2L$-periodic function, and state the orthogonality relations that justify them.
3. **(Odd function series)** Find the Fourier series of the 2π-periodic extension of $f(x) = x$ on $-\pi \le x \le \pi$, computing b_n in full.
4. **(Symmetry shortcut)** Without integrating, explain why $f(x) = x^2$ on $-\pi \le x \le \pi$ has $b_n = 0$ for all n, and which series type it has.
5. **(Half-range sine)** Expand $f(x) = 1$ on $0 \le x \le L$ in a half-range sine series, giving b_n and the resulting series.
6. **(Separation of variables)** Separate $u_t = \alpha^2 u_{xx}$ via $u = X(x)T(t)$; state the two ODEs and, with $u(0,t) = u(L,t) = 0$, the spatial eigenpairs and time factor.
7. **(Laplacian / harmonic)** Determine whether $u(x,y) = x^2 - y^2$ and $v(x,y) = e^x \sin y$ are harmonic by computing ∇^2 of each.
8. **(Classification)** Classify $u_t = \alpha^2 u_{xx}$, $u_{tt} = c^2 u_{xx}$, and $u_{xx} + u_{yy} = 0$ as parabolic, hyperbolic, or elliptic, and name the phenomenon each models.
9. **(Heat equation capstone)** Solve $u_t = \alpha^2 u_{xx}$ on $0 \le x \le L$, $u(0,t) = u(L,t) = 0$, $u(x,0) = 100$, computing the coefficients in full.
10. **(Wave equation)** Use d'Alembert's formula to solve $u_{tt} = c^2 u_{xx}$ on the line with $u(x,0) = f(x)$ and $u_t(x,0) = 0$, and describe the motion.

Part III — Complete Solutions

Solution 1. Match $f = \frac{a_0}{2} + \sum_n (a_n \cos n\pi x + b_n \sin n\pi x)$ with $L = 1$. The constant $\frac{a_0}{2} = 4$, so

$$a_0 = 8; \qquad a_2 = -3 \text{ (from } -3\cos 2\pi x\text{)}; \qquad b_1 = 1 \text{ (from } \sin \pi x\text{)}.$$

All other coefficients are 0.

Solution 2. For a $2L$-periodic f,

$$a_n = \frac{1}{L}\int_{-L}^{L} f \cos\frac{n\pi x}{L}\,dx, \qquad b_n = \frac{1}{L}\int_{-L}^{L} f \sin\frac{n\pi x}{L}\,dx.$$

They follow from orthogonality: $\int_{-L}^{L} \cos\frac{n\pi x}{L}\cos\frac{m\pi x}{L}dx = L\delta_{nm}$, the same for sines, and $\int_{-L}^{L}\cos\frac{n\pi x}{L}\sin\frac{m\pi x}{L}dx = 0$ — integrating f against one basis wave isolates its coefficient.

Solution 3. $f(x) = x$ is odd, so $a_n = 0$ and only b_n survive. With $L = \pi$, integration by parts gives

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x\sin(nx)\,dx = \frac{2}{\pi}\Big[-\frac{\pi}{n}(-1)^n\Big] = (-1)^{n+1}\frac{2}{n}.$$

Hence $x = \sum_{n\geq 1}(-1)^{n+1}\frac{2}{n}\sin(nx) = 2(\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots)$.

Solution 4. $f(x) = x^2$ is *even* ($f(-x) = f(x)$). The integrand $x^2\sin\frac{n\pi x}{L}$ is then odd, so

$$b_n = \frac{1}{L}\int_{-L}^{L} x^2 \sin\frac{n\pi x}{L}\,dx = 0 \quad \text{(odd integrand over a symmetric interval).}$$

It therefore has a pure *cosine* series (only a_0 and a_n).

Solution 5. Half-range sine: $b_n = \frac{2}{L}\int_0^L \sin\frac{n\pi x}{L}\,dx$. Then

$$b_n = \frac{2}{L}\cdot\frac{L}{n\pi}\big(1-(-1)^n\big) = \frac{2}{n\pi}\big(1-(-1)^n\big) = \frac{4}{n\pi}\ (n \text{ odd}),\ 0\ (n \text{ even}).$$

So $1 = \frac{4}{\pi}\sum_{k\geq 0}\frac{1}{2k+1}\sin\frac{(2k+1)\pi x}{L}$ on $0 < x < L$.

Solution 6. Substituting $u = XT$ into $u_t = \alpha^2 u_{xx}$ and dividing by $\alpha^2 XT$ gives $\frac{T'}{\alpha^2 T} = \frac{X''}{X} = -\lambda$. Thus

$$X'' + \lambda X = 0,\ X(0) = X(L) = 0 \ \Rightarrow\ \lambda_n = \left(\tfrac{n\pi}{L}\right)^2,\ X_n = \sin\tfrac{n\pi x}{L};$$
$$T' + \alpha^2\lambda_n T = 0 \ \Rightarrow\ T_n = e^{-\alpha^2(n\pi/L)^2 t}.$$

Solution 7. For $u = x^2 - y^2$: $u_{xx} = 2$, $u_{yy} = -2$, so

$$\nabla^2 u = 2 - 2 = 0 \text{ (harmonic).}$$

For $v = e^x\sin y$: $v_{xx} = e^x \sin y$, $v_{yy} = -e^x\sin y$, so

$$\nabla^2 v = e^x\sin y - e^x \sin y = 0 \text{ (also harmonic).}$$

Solution 8. By the second-order classification:

$$u_t = \alpha^2 u_{xx}: \text{ parabolic (diffusion/heat),}$$
$$u_{tt} = c^2 u_{xx}: \text{ hyperbolic (vibration/waves),}$$
$$u_{xx} + u_{yy} = 0: \text{ elliptic (steady state).}$$

Solution 9. Separation gives $X_n = \sin\frac{n\pi x}{L}$, $T_n = e^{-\alpha^2(n\pi/L)^2 t}$, so $u = \sum_n b_n \sin\frac{n\pi x}{L} e^{-\alpha^2(n\pi/L)^2 t}$. Matching $u(x,0) = 100$,

$$b_n = \frac{2}{L}\int_0^L 100 \sin\frac{n\pi x}{L}\,dx = \frac{200}{n\pi}\big(1-(-1)^n\big) = \frac{400}{n\pi}\ (n \text{ odd}),\ 0\ (n \text{ even}).$$

$$u(x,t) = \frac{400}{\pi}\sum_{k\ge 0}\frac{1}{2k+1}\sin\frac{(2k+1)\pi x}{L}\,e^{-\alpha^2((2k+1)\pi/L)^2 t} \xrightarrow{t\to\infty} 0.$$

Solution 10. With initial velocity $g \equiv 0$, d'Alembert's formula reduces to

$$u(x,t) = \tfrac{1}{2}\big[f(x-ct) + f(x+ct)\big].$$

The initial shape splits into two half-amplitude copies, one traveling right at speed c and one left at speed c; their sum reconstructs f at $t = 0$ and separates thereafter.

Unit Key Takeaway

One idea organizes the whole unit: decompose into frequencies, then solve frequency by frequency. Fourier series rebuild a function from orthogonal sines and cosines, with symmetry choosing the cosine or sine form and half-range extensions adapting it to a boundary. For PDEs, separation of variables turns the equation into the Unit 17 eigenvalue BVP plus a time ODE, superposition assembles the modes, and the initial condition is fitted by a Fourier series. The Laplacian is the shared spatial operator behind the heat, wave, and Laplace equations — diffusion that smooths, vibration that propagates, and the harmonic steady state.